



Kabul University
Computer Science Faculty
Information Technology Department

Computer Architecture

Computer Evolution and Performance

Contents

- Purpose of this subject
- What we will cover
- Evolution of computer system
- Computer organization/Computer architecture
- History of computer system

Purpose

- How a computer actually work?
- How data and program are storing in computer?
- How we can make computer faster?

The topics to be cover

- ALU
- Instruction set architecture
- Input/output design
- Memory unit design
- Parallel processing
- Pip line
- Processor design

Evolution of computer system

- Computer is a part and parcel of our daily lives.
 - They are every where
 - Laptops, mobile phones, tablets and so on.
- It is required to understand how a computer work.
 - What are there inside the computer?
 - How there inside components work?
- The difference between **computer architecture** and *computer organization*.

Computer organization/Computer architecture

- Computer organization
 - Design of the components and functional blocks using which computer systems are built.
 - For example: in computer system we have functional blocks such as processor inside processor we have many parts like ALU, Register and so on, we use adder in ALU so which kinds of adder must be used, this is the focuses area of computer organization.
- Computer architecture
 - How to integrate the computer components to achieve the desired level of performance.

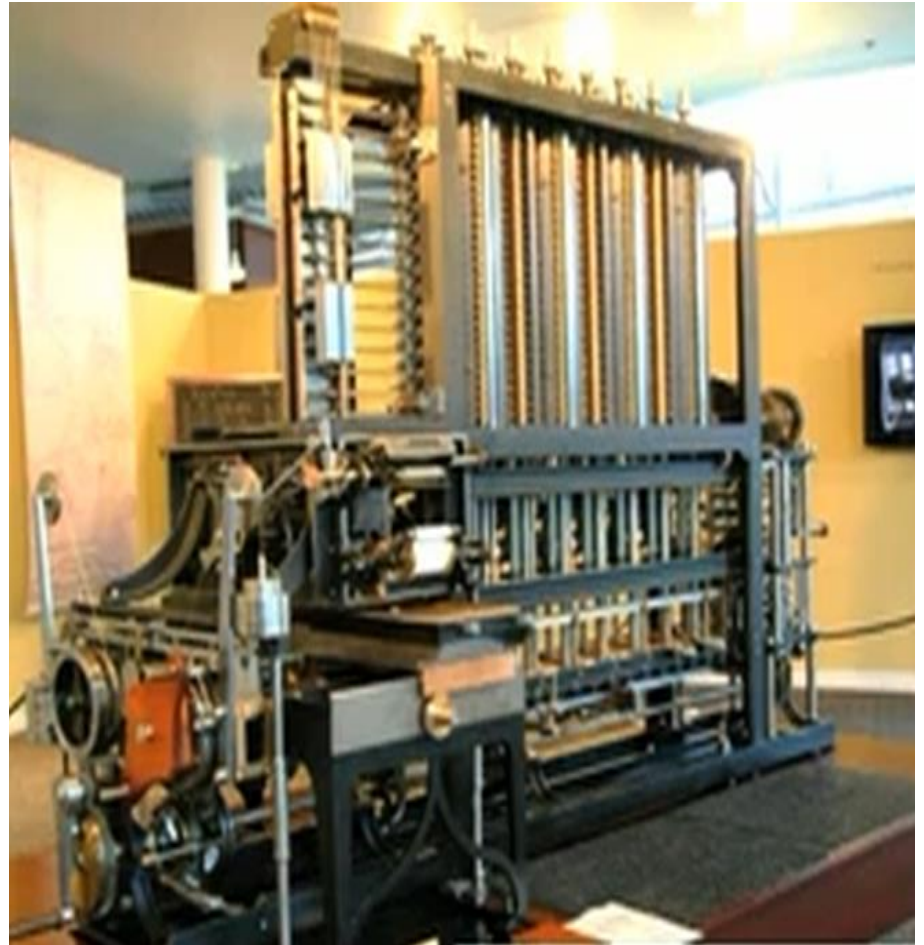
PASCALINE 1642

- Mechanical calculator invented by B. Pascal.
- Could add and subtract only two numbers directly.



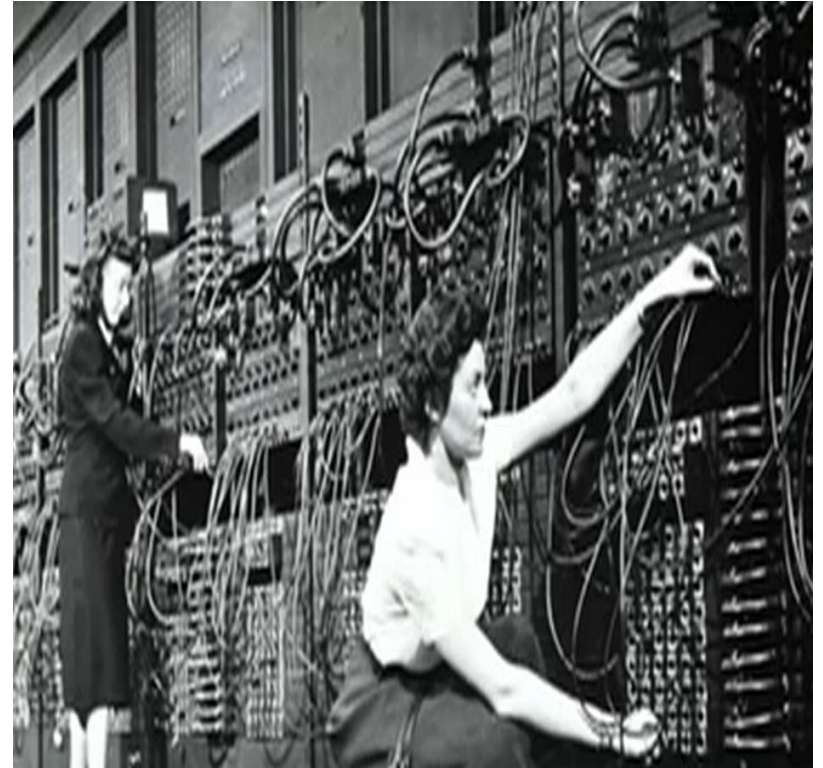
Babbage Engine

- First automatic **computing engine**.
- Designed by **Charles Babbage** in 19th century, but he could not build it.
- The first complete **Babbage engine** was built in 2002, 153 years after it was designed.
 - 8000 parts.
 - Weighed 5 tons.
 - 11 feet in length.



ENIAC

- Electrical numerical integrator and calculator
- Developed at the **Pennsylvania university.**
- Used 18,000 **vacuum tubes.**
 - Weighed 30 tons
 - Occupied a 30ft * 50ft spaces.



Harvard Mark 1

- Built at the Harvard University in 1944 with support from IBM.
- Used mechanical relays(switches) to represent data.
- It weighed 35 tons, and required 500 miles of wiring.



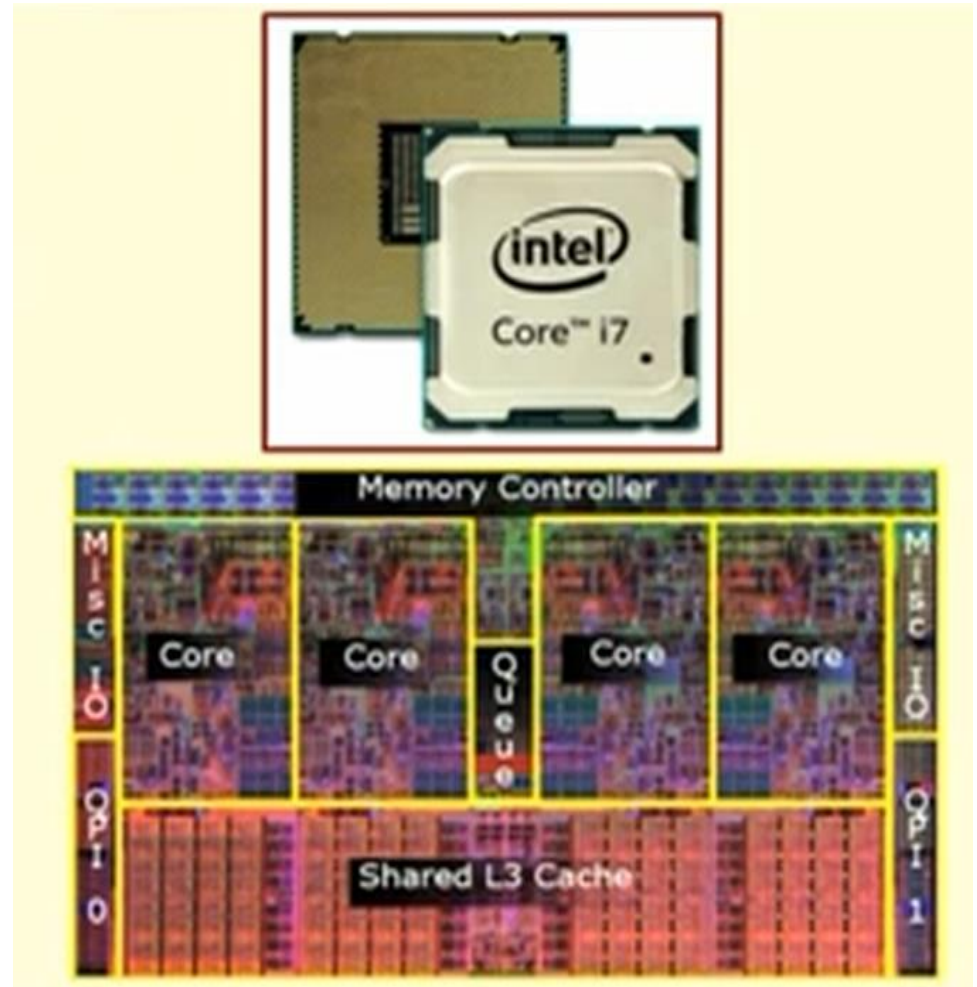
IBM System/360

- Very popular mainframe computer of the 60s and 70s.
- Introduced many advanced architecture concepts that appears in microprocessors several decades later.



Intel Core i7

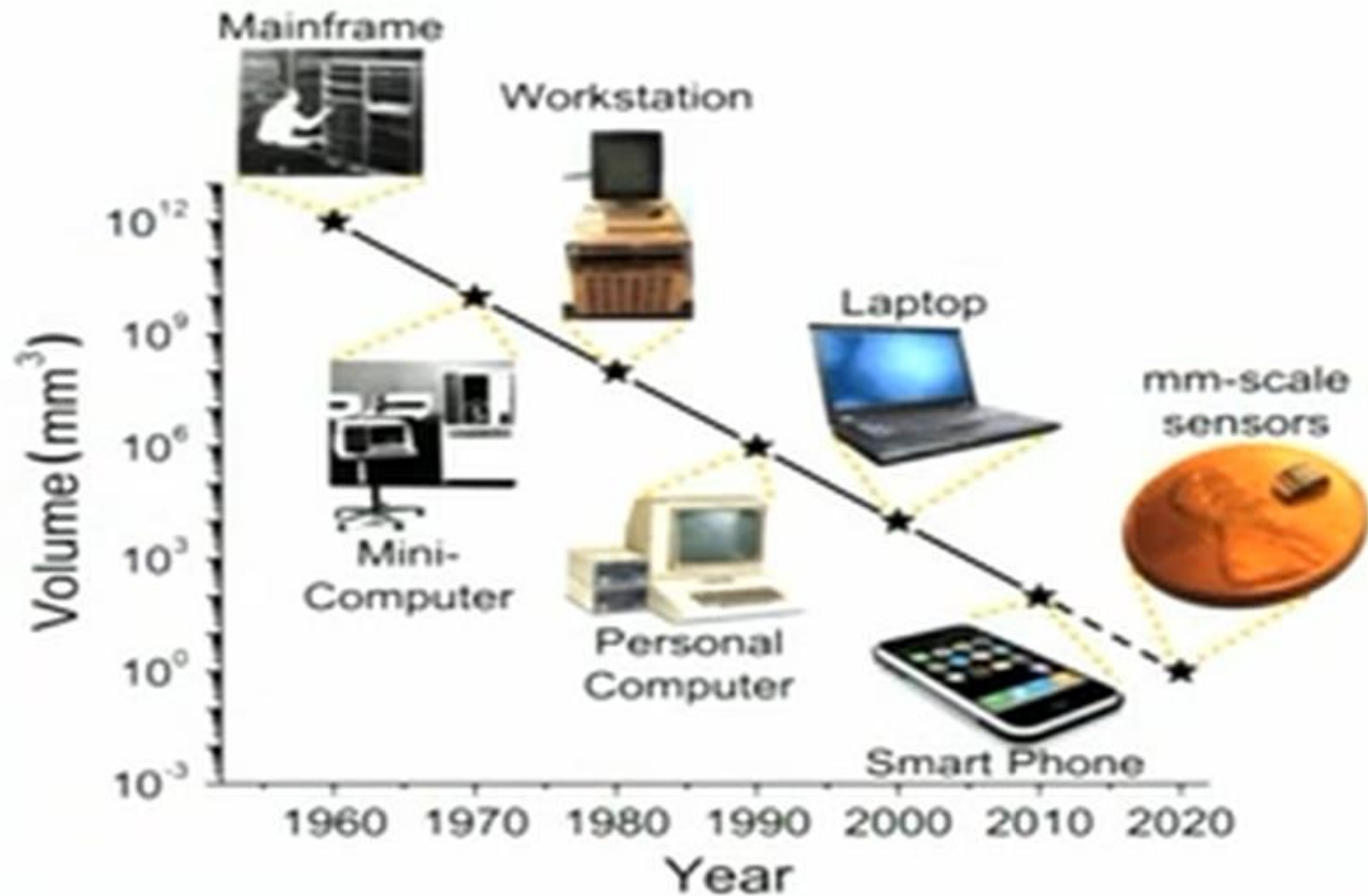
- A modern processor chip, that comes in dual-core, quad-core and 6-core variants.
- 64-bit processor that comes with various microarchitectures.



Generation of computers

Generation	Main Technology	Representative Systems
First (1945-54)	Vacuum tubes, relays	Machine & assembly language ENIAC, IBM-701
Second (1955-64)	Transistors, memories, I/O processors	Batch processing systems, HLL IBM-7090
Third (1965-74)	SSI and MSI integrated circuits Microprogramming	Multiprogramming / Time sharing IBM 360, Intel 8008
Fourth (1975-84)	LSI and VLSI integrated circuits	Multiprocessors Intel 8086, 8088
Fifth (1984-90)	VLSI, multiprocessor on-chip	Parallel computing, Intel 486
Sixth (1990 onwards)	ULSI, scalable architecture, post-CMOS technologies	Massively parallel processors Pentium, SUN Ultra workstations

Evolution of types of computer



Evolution of PC form factors over the years



Standard-ATX



Micro-ATX



Mini-ITX



Nano-ITX



Pico-ITX



Inside a laptop

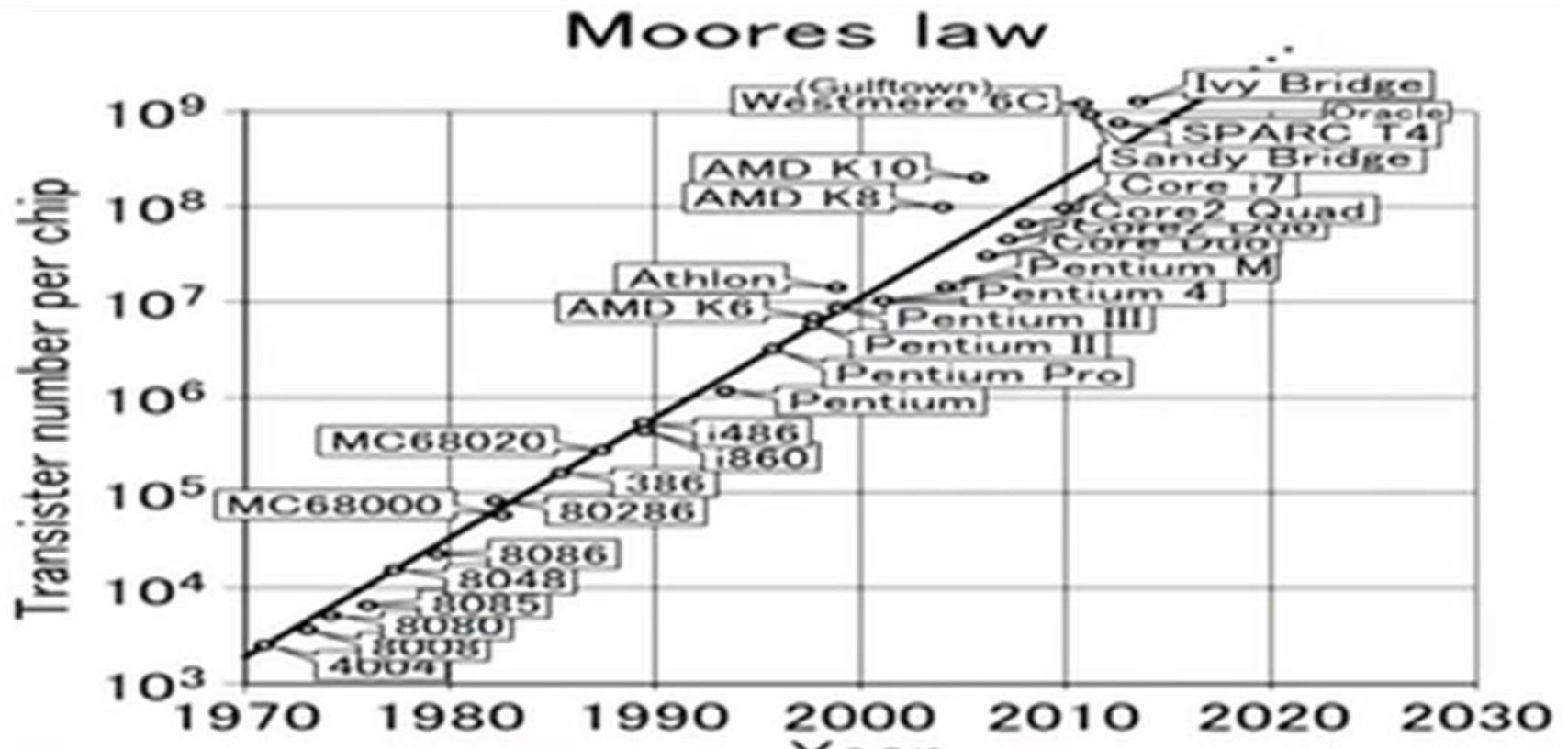
- Miniaturization in feature sizes of all parts.
- Hard drive getting replaced by flash-based memory devices.
- Cooling is a major issues.



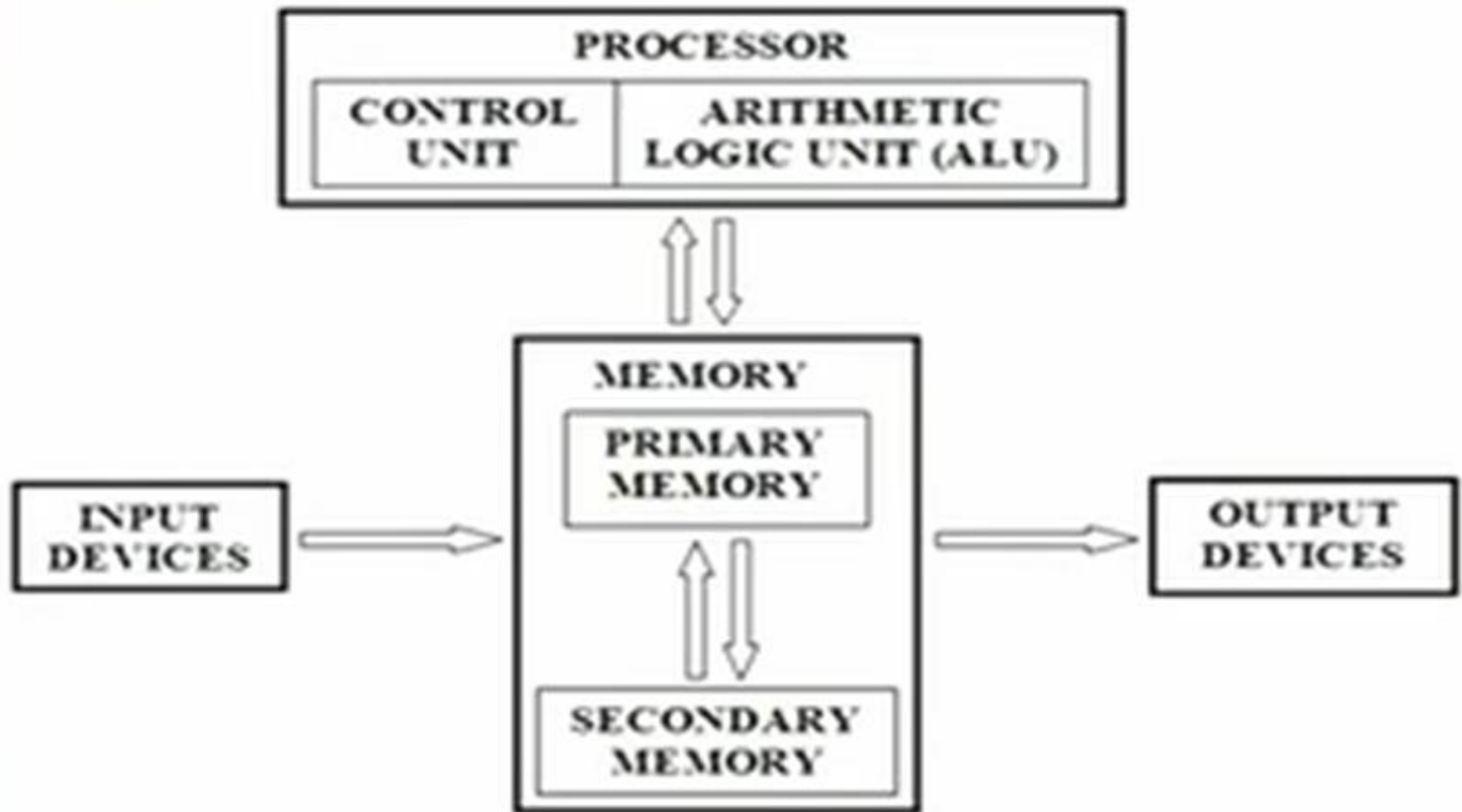
Moore law

- Refers to an observation made by Intel co-founder Gordon Moore in 1965. he noticed that the number of transistors per square inch on integrated circuits had doubled every year since their invention

Transistor numbers per chip



Simplified Block Diagram of a Computer System



Cont.

- All instructions and data are stored in memory.
- An instruction and the required data are brought into the processor for execution.
- Input and output devices interfaced with the outside world.

Inside the Processor

- Processor is also called central processing unit(CPU)
 - Consists of a Control unit and **arithmetic logic unite**(ALU).
 - All calculations happen inside the ALU.
 - The **control unit** generates sequence of control signals to carry out all operations.
 - The processor fetches and instruction from memory for execution.
 - An instruction specifies the exact operation to be carried out.
 - It also specifies the data that are to be operated on.
 - A program refers to a set of instructions that are required to carry out a specific task.(ex: sorting as set of numbers).

The role of ALU

- It contains several registers, some general purpose and some special purpose, for temporary storage of data.
- It contains circuitry to carry out logic operations, like AND, OR, NOT, SHIFT, COMPARE and so on.
- It contains circuitry to carry out arithmetic operations like addition, subtraction, multiplication, division and so on.
- During the instruction execution the data(operands) are brought in and store in some register after operation carried out, and the result are stored back to some register or memory.

The role of control unit

- It generate the signals to control the states of various functional units and send control signal to control their states.
- To carry out a specific operation ($R1 \leftarrow R2+R3$), the control unite must generate control signals in a specific sequence.
 - Enable the outputs of registers R2 and R3.
 - Select the addition operation.
 - Store the output of the adder circuit into register R1.
- When an instruction is fetched from memory, the operation (called opcode) is decoded by the control unit, and control signals issued.

Cont.

- Example:
- `ADD A,B`
- `ADD R1,R2`
- `ADD` is called opcode, and `R1,R2` is called operand in which the operation to be done

Memory unit

- Two main types of memory subsystems.
 - Primary or main memory
 - Secondary memory
- The processor only has direct access to primary memory
- In reality, the memory system is implemented as hierarchy of several levels.
 - L1 cache, L2 cache, L3 cache, primary memory, secondary memory
 - Objective is to provide faster memory access.

Various different types of memory

- Random Access Memory(RAM)
- Read Only Memory (ROM)
- Magnetic Disk
- Flash Memory
- ect

Cont.



Input Unit

- Used to feed data to the computer system from the external environment.
 - Data are transferred to the processor/memory after appropriated encoding.
- Common input devices:
 - Keyboard
 - Mouse
 - Camera
 - And so on

Cont.



Output Unit

- Used to send the result of some computation to the outside world.
- Common output devices:
 - LCD/LED screen
 - Printer
 - Speaker
 - Projection system

Cont.



Thanks for attention